NET (BMEVITMAC09): two mock midterms at the level of the 2026 slides

Written by us, not official papers. Each question is the twin of an exercise or worked example actually shown in the 2026 slides (same type, same number of steps, same size of numbers), with new numbers; course questions stay at the level of detail of the slides. One question per session, sessions 1 to 6 in order. Format as announced by the lecturer and reported by Elisha: 6 open questions, 30 points, pass mark 12/30. Calculations checked by `calc_sujets_slides_net.py`.

Exercises and worked examples in the 2026 slides

Slide, page	Example	Difficulty
hta1 p. 13	PCM: 8 kHz × 8 bits = 64 kbit/s	1 multiplication
hta1p p. 18, 19	Token and leaky bucket design: K = average, B = (max −avg)× burst	formula, no numbers
hta1p p. 20	Statistical multiplexing: $N = 500$, $p = 1\%$, mean 5, Chebyshev bound 1/9	3 steps, small numbers
hta1p p. 24 to 26	Same record in XML, JSON, YAML; “Possible exam question”	2 scalars and 1 list of strings
hta2p p. 4, 7, 8	Class table, hosts per prefix ($2^{32-n} - 2$), private ranges	table lookup
hta2p p. 11	Ex. 1: netmask to prefix (255.224.0.0 = /11), /19 to netmask, invalid mask	1 step each
hta2p p. 12	Ex. 2: which of 5 addresses share 152.66.163.9/18 (binary)	5 binary comparisons
hta2p p. 13	Ex. 3: 3-line table, 1 destination, longest prefix match	binary, 1 destination
hta2p p. 14	Ex. 4: 4 lines on one interface merged into a /9, one misaligned line	2 merges, 1 alignment check
hta3 p. 10, 12; hta3p p. 10 to 20	DORA diagram; URL parts; ifconfig, ip, route examples; ping, traceroute	recall, command syntax
hta4p p. 5, 6	Distance vector on a line of 6 nodes; count to infinity after one node fails	3 to 5 rounds
hta4p p. 8 to 14	Dijkstra from C on 7 nodes, 9 links, queue shown per round	6 rounds
hta5 p. 9	3-way handshake: SYN (Sc), SYN,ACK (Ss, Sc+1), ACK (Sc+1, Ss+1)	symbolic
hta5 p. 18, 19	NAT: client, gateway inside and outside addresses, server	address rewriting
hta5p p. 8, 9	cwnd 1, 2, 4, 8 then 9, 10 with segment numbers (1; 2, 3; 4 to 7…)	counting per round
hta5p p. 13	Fast retransmit: segments 20 to 26, 21 lost, ACKs 21, 21, 21, 21, 21, 26	1 sequence
hta5p p. 16	16-bit window, window scale shift up to 14 bits, 1 GB	1 step
hta6 p. 8; hta6p p. 7 to 9	HTTP methods and code classes; SMTP, POP3, IMAP ports and commands	recall

Compared with the 2018 midterm, these papers drop what the slides never compute: VLSM allocation, aggregation with non-contiguous traps, router interface plans with ARP traces, Reno over many rounds with a capping convention, bandwidth-delay products, fragmentation tables.

Paper 1

6 questions, one per session, 5 points each, pass mark 12/30.

1. Session 1: XML to JSON and YAML (5 points)

mirrors hta1p p. 24 to 26 (“Possible exam question”)

```
<Router>
  <id>7</id>
  <name>Edge Router</name>
  <interfaces>
    <ifname>eth0</ifname>
    <ifname>eth1</ifname>
    <ifname>wlan0</ifname>
  </interfaces>
</Router>
```

(a) Turn the given XML into JSON. (2 pts)
(b) Turn it into YAML. (2 pts)
(c) Why was YAML introduced next to JSON, and what is it mainly used for? (1 pt)

2. Session 2: netmasks and same subnet (5 points)

mirrors hta2p p. 11 (Example 1) and p. 12 (Example 2)

(a) Prefix length of netmask 255.255.192.0? Netmask of /20? Is 255.255.240.128 valid? (1.5 pts)

(b) Which addresses are on the same subnet as 130.88.201.7/19? Show the 3rd octet in binary. 1) 130.88.215.40 2) 130.88.190.3 3) 131.88.201.9 4) 130.88.224.5 5) 130.88.192.12 6) 130.88.207.300 (3.5 pts)

3. Session 3: DHCP and configuration commands (5 points)

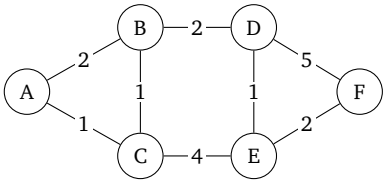
mirrors hta3 p. 10 (DORA diagram); hta3p p. 14 (iproute2 examples)

(a) Name the four DHCP messages in order. Which are broadcast, and why? Which transport and ports does DHCP use? (2.5 pts)

(b) With the `ip` command: give eth1 the address 172.16.5.20/24, bring it up, and add a route to 10.40.0.0/16 via 172.16.5.1. (2.5 pts)

4. Session 4: Dijkstra's algorithm (5 points)

mirrors hta4p p. 8 to 14 (LS example, run Dijkstra)



Our node is A.

(a) As on the slides, give for each round the node taken from the priority queue, the new path costs and the queue. (3.5 pts)

(b) Final cost and next hop from A for every node. (1.5 pts)

5. Session 5: slow start and congestion avoidance (5 points)

mirrors hta5p p. 8 and 9 (cwnd and segment numbers per round)

As on the slides: slow start doubles cwnd each round until cwnd reaches ssthresh; congestion avoidance adds 1 per round with ssthresh = cwnd; on a timeout ssthresh is halved and slow start restarts from cwnd = 1. Start: cwnd = 1, ssthresh = 4, segments numbered from 1.

(a) Give cwnd and the segment numbers sent in rounds 1 to 7. (3 pts)

(b) A timeout happens at the end of round 7. ssthresh, and cwnd in rounds 8 to 11? (2 pts)

6. Session 6: HTTP (5 points)

mirrors hta6 p. 8 to 11 (methods, response codes, versions, state)

(a) Which method: get a page; send a search term; get only the metadata of an object? (1.5 pts)

(b) What do the classes 2XX, 3XX, 4XX and 5XX mean? Give one code of each from the slides. (1.5 pts)

(c) What did HTTP/1.1, HTTP/2 and HTTP/3 each bring? How does a stateless protocol keep a shopping cart? (2 pts)

1. Session 1: PCM and statistical multiplexing (5 points)

mirrors hta1 p. 13 (PCM); hta1p p. 20 (statistical multiplexing)

(a) A wideband voice service keeps audio up to 8 kHz and quantizes each sample to 8 bits. Sampling rate and bitrate? (1.5 pts)

(b) $N = 300$ users, each using the network with probability $p = 2\%$. Average number of active users? Which distribution, and which approximation for large N and small p? Apply Chebyshev's inequality with $k = 3$ as on the slide: which L, and which bound on the blocking probability? (3.5 pts)

2. Session 2: longest prefix match and table compression (5 points)

mirrors hta2p p. 13 (Example 3) and p. 14 (Example 4)

(a) Table: 98.32.0.0/11 eth1; 98.48.0.0/12 eth4; 98.56.0.0/13 eth2. Outgoing interface for 98.52.130.9? Show the 2nd octets in binary. (2 pts)

(b) All four lines go to eth3: 200.17.36.0/21; 200.17.48.0/21; 200.17.56.0/22; 200.17.60.0/22. Which lines can be merged, into what? Which line is incorrectly given? (3 pts)

3. Session 3: URL, DNS and traceroute (5 points)

mirrors hta3 p. 6, 12, 14, 15; hta3p p. 18 to 20

(a) Split `https://www.example.org/docs/net/intro.html` into scheme, location, path and file. Which part does DNS resolve? (1.5 pts)

(b) What are the A, AAAA and MX records? How does a client know its local DNS resolver? (1.5 pts)

(c) How does traceroute find the routers on the path? Which ICMP message does it use? (2 pts)

4. Session 4: distance vector and count to infinity (5 points)

mirrors hta4p p. 5 (DV example) and p. 6 (count to infinity)

Five routers on a line: P, Q, R, S, T (hop count). Initially nothing is known.

(a) After how many rounds does P learn about T? What does R advertise once it knows all nodes within 2 hops? (1.5 pts)

(b) Q goes down. R detects it and marks P and Q unreachable. Then R receives S's update "P in 3 hops". Give the next three updates for destination P. (2.5 pts)

(c) How would split horizon and poisoned reverse prevent this? (1 pt)

5. Session 5: TCP handshake and fast retransmit (5 points)

mirrors hta5 p. 9 (3-way handshake); hta5p p. 13 (fast retransmit)

(a) The client picks Sc = 4100, the server Ss = 900. Give flags, Sequence and Acknowledgement of the three handshake segments. (2 pts)

(b) As on the slide, segments are numbered one by one. The sender sends 70 to 75; segment 72 is lost. Give the ACK the receiver returns for each segment, when Reno retransmits, and the ACK after 72 arrives. (3 pts)

6. Session 6: e-mail (5 points)

mirrors hta6p p. 4 and 7 to 9 (architecture, SMTP, POP3, IMAP)

(a) A message goes from Alice's mail client to Bob's mailbox and Bob reads it. Name the agents in order and the protocol at each step. How does the sending MTA find Bob's domain's MTA? (2.5 pts)

(b) Give the ports of SMTP (server to server and submission), POP3 and IMAP. (1 pt)

(c) Why is IMAP better than POP3 for reading mail on two devices? (1.5 pts)

Solutions

Paper 1: solutions

6 questions, one per session, 5 points each, pass mark 12/30.

1. Session 1: XML to JSON and YAML (5 points)

mirrors hta1p p. 24 to 26 ("Possible exam question")

```
<Router>
  <id>7</id>
  <name>Edge Router</name>
  <interfaces>
    <ifname>eth0</ifname>
    <ifname>eth1</ifname>
    <ifname>wlan0</ifname>
  </interfaces>
</Router>
```

(a) Turn the given XML into JSON. (2 pts)
(b) Turn it into YAML. (2 pts)
(c) Why was YAML introduced next to JSON, and what is it mainly used for? (1 pt)

```
{
  "id": 7,
  "name": "Edge Router",
  "interfaces": [ "eth0", "eth1",
                  "wlan0" ]
}
```

```
id: 7
name: Edge Router
interfaces:
  - eth0
  - eth1
  - wlan0
```

(c) JSON is verbose with braces and brackets; YAML uses indentation like Python. Mainly for configuration files.

Marking. (a) object 1, array 0.5, 7 without quotes 0.5. (b) keys 1, hyphenated list 1. (c) 1. **Pitfall.** Keeping `ifname` inside the list. As on the slide, the root's content becomes the top-level object (an outer `"Router"` key is also consistent).

2. Session 2: netmasks and same subnet (5 points)

mirrors hta2p p. 11 (Example 1) and p. 12 (Example 2)

(a) Prefix length of netmask 255.255.192.0? Netmask of /20? Is 255.255.240.128 valid? (1.5 pts)

192 = 11000000: /18. 255.255.240.0. Invalid: all 1s must come before all 0s.

(b) Which addresses are on the same subnet as 130.88.201.7/19? Show the 3rd octet in binary. 1) 130.88.215.40 2) 130.88.190.3 3) 131.88.201.9 4) 130.88.224.5 5) 130.88.192.12 6) 130.88.207.300 (3.5 pts)

/19 = 8 + 8 + 3: compare the first 3 bits of the 3rd octet; 201 = 11001001. 215 = 11010111 **yes**; 190 = 10111110 no; 131.x: 1st octet differs, no; 224 = 11100000 no; 192 = 11000000 **yes**; 300 > 255: invalid.

Marking. (a) 0.5 each. (b) about 0.6 per address.

3. Session 3: DHCP and configuration commands (5 points)

mirrors hta3 p. 10 (DORA diagram); hta3p p. 14 (iproute2 examples)

(a) Name the four DHCP messages in order. Which are broadcast, and why? Which transport and ports does DHCP use? (2.5 pts)

Discover, Offer, Request, Acknowledgement (DORA). Discover is broadcast (the client has no address and does not know the server); Request is also broadcast, to notify all DHCP servers. UDP, server port 67, client port 68.

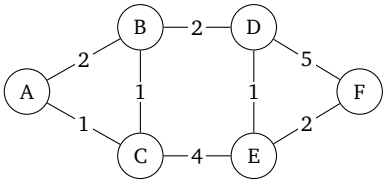
(b) With the `ip` command: give eth1 the address 172.16.5.20/24, bring it up, and add a route to 10.40.0.0/16 via 172.16.5.1. (2.5 pts)

`ip addr add 172.16.5.20/24 dev eth1;` `ip link set dev eth1 up;` `ip route add 10.40.0.0/16 dev eth1 via 172.16.5.1.`

Marking. (a) order 1, broadcast 1, ports 0.5. (b) about 0.8 per command; the older `ifconfig` and `route` forms earn partial credit.

4. Session 4: Dijkstra's algorithm (5 points)

mirrors hta4p p. 8 to 14 (LS example, run Dijkstra)



Our node is A.
(a) As on the slides, give for each round the node taken from the priority queue, the new path costs and the queue. (3.5 pts)
(b) Final cost and next hop from A for every node. (1.5 pts)

Round	Taken	New paths; queue after the round
1	A (0)	A-B 2, A-C 1; queue C 1, B 2
2	C (1)	A-C-B 2 (not better), A-C-E 5; queue B 2, E 5
3	B (2)	A-B-D 4; queue D 4, E 5
4	D (4)	A-B-D-E 5 (not better), A-B-D-F 9; queue E 5, F 9
5	E (5)	A-C-E-F 7 (better); queue F 7
6	F (7)	done

(b) B 2 via B; C 1 via C; D 4 via B; E 5 via C; F 7 via C.

Marking. (a) about 0.6 per round. (b) 1.5. **Pitfall.** Keeping F at 9.

5. Session 5: slow start and congestion avoidance (5 points)

mirrors hta5p p. 8 and 9 (cwnd and segment numbers per round)

As on the slides: slow start doubles cwnd each round until cwnd reaches ssthresh; congestion avoidance adds 1 per round with ssthresh = cwnd; on a timeout ssthresh is halved and slow start restarts from cwnd = 1. Start: cwnd = 1, ssthresh = 4, segments numbered from 1.

(a) Give cwnd and the segment numbers sent in rounds 1 to 7. (3 pts)

cwnd 1, 2, 4, 5, 6, 7, 8; segments 1; 2, 3; 4 to 7; 8 to 12; 13 to 18; 19 to 25; 26 to 33.

(b) A timeout happens at the end of round 7. ssthresh, and cwnd in rounds 8 to 11? (2 pts)

ssthresh = cwnd = 8 before, halved to **4**; cwnd 1, 2, 4, 5.

Marking. (a) cwnd 1.5, segments 1.5. (b) 1 and 1. **Pitfall.** Doubling 4 to 8 in round 4.

6. Session 6: HTTP (5 points)

mirrors hta6 p. 8 to 11 (methods, response codes, versions, state)

(a) Which method: get a page; send a search term; get only the metadata of an object? (1.5 pts)

GET; POST; HEAD.

(b) What do the classes 2XX, 3XX, 4XX and 5XX mean? Give one code of each from the slides. (1.5 pts)

Success (200 OK); redirection (301 Moved Permanently); client error (403, 404 Not Found); server error (500, 502, 504).

(c) What did HTTP/1.1, HTTP/2 and HTTP/3 each bring? How does a stateless protocol keep a shopping cart? (2 pts)

1.1: reuse of the TCP connection for several objects. 2: header compression, request multiplexing. 3: QUIC instead of TCP. Cookies: the response sets a key-value pair, the browser sends it back with later requests to the same site.

Paper 2: solutions

6 questions, one per session, 5 points each, pass mark 12/30.

1. Session 1: PCM and statistical multiplexing (5 points)

mirrors hta1 p. 13 (PCM); hta1p p. 20 (statistical multiplexing)

(a) A wideband voice service keeps audio up to 8 kHz and quantizes each sample to 8 bits. Sampling rate and bitrate? (1.5 pts)

Nyquist: twice the maximum frequency, 16 kHz; $16{,}000 \times 8 =$ **128 kbit/s**.

(b) $N = 300$ users, each using the network with probability $p = 2\%$. Average number of active users? Which distribution, and which approximation for large N and small p? Apply Chebyshev's inequality with $k = 3$ as on the slide: which L, and which bound on the blocking probability? (3.5 pts)

$Np =$ **6**. Binomial, approximated by Poisson with $\lambda = Np = 6$. As written on the slide ($\sigma = \lambda$): $L = 6 + 3 \times 6 =$ **24**, $\Pr(|X - 6| \geq 18) \leq 1/9$. With the true standard deviation $\sigma = \sqrt{6} \approx 2.45$: $L = 6 + 7.35 = 13.35$, so 14 links, same bound 1/9. Chebyshev does not depend on the distribution, so it overestimates.

Marking. (a) 0.75 each. (b) mean 1, distributions 1, L and bound 1.5; either σ convention accepted if stated.

2. Session 2: longest prefix match and table compression (5 points)

mirrors hta2p p. 13 (Example 3) and p. 14 (Example 4)

(a) Table: 98.32.0.0/11 eth1; 98.48.0.0/12 eth4; 98.56.0.0/13 eth2. Outgoing interface for 98.52.130.9? Show the 2nd octets in binary. (2 pts)

32 = `00100000` (/11 fixes 3 bits), 48 = `00110000` (4 bits), 56 = `00111000` (5 bits); 52 = `00110100` matches lines 1 and 2, not 3: longest wins, **eth4**.

(b) All four lines go to eth3: 200.17.36.0/21; 200.17.48.0/21; 200.17.56.0/22; 200.17.60.0/22. Which lines can be merged, into what? Which line is incorrectly given? (3 pts)

56/22 (56 to 59) + 60/22 (60 to 63) = 56/21 (only the last prefix bit differs); + 48/21 (48 to 55) = **200.17.48.0/20** (48 to 63). Line 1: in a /21 the 3rd octet is a multiple of 8, 36 is not; it really is 200.17.32.0/21 (32 to 39), and 40 to 47 are missing, so it cannot be merged.

Marking. (a) binary 1, result 1. (b) first merge 1, /20 1, misaligned line 1.

3. Session 3: URL, DNS and traceroute (5 points)

mirrors hta3 p. 6, 12, 14, 15; hta3p p. 18 to 20

(a) Split `https://www.example.org/docs/net/intro.html` into scheme, location, path and file. Which part does DNS resolve? (1.5 pts)

Scheme `https`; location `www.example.org` (resolved to an IP address by DNS); path `/docs/net`; file `intro.html`.

(b) What are the A, AAAA and MX records? How does a client know its local DNS resolver? (1.5 pts)

IPv4 address; IPv6 address; e-mail server. Typically from DHCP.

(c) How does traceroute find the routers on the path? Which ICMP message does it use? (2 pts)

It sends packets with TTL = 1, 2, 3 ... ; the router where the TTL expires sends back ICMP Time Exceeded (type 11), which reveals it. Probes: UDP to a random port by default, ICMP Echo Request with `-I`, TCP SYN with `-T`.

4. Session 4: distance vector and count to infinity (5 points)

mirrors hta4p p. 5 (DV example) and p. 6 (count to infinity)

Five routers on a line: P, Q, R, S, T (hop count). Initially nothing is known.

(a) After how many rounds does P learn about T? What does R advertise once it knows all nodes within 2 hops? (1.5 pts)

T is 4 hops away: after **4 rounds**. R says [[P, 2], [Q, 1], [S, 1], [T, 2]], without next hops.

(b) Q goes down. R detects it and marks P and Q unreachable. Then R receives S's update "P in 3 hops". Give the next three updates for destination P. (2.5 pts)

R does not know it is S's next hop: R sets P to **4 via S**. R says 4 to S; R is S's next hop towards P, so S updates to **5**. S says 5: R updates to **6**. And so on, counting to infinity.

(c) How would split horizon and poisoned reverse prevent this? (1 pt)

Split horizon: S does not advertise P to R, its next hop. Poisoned reverse: S advertises P to R with an infinite cost, so R does not choose S as next hop.

5. Session 5: TCP handshake and fast retransmit (5 points)

mirrors hta5 p. 9 (3-way handshake); hta5p p. 13 (fast retransmit)

(a) The client picks Sc = 4100, the server Ss = 900. Give flags, Sequence and Acknowledgement of the three handshake segments. (2 pts)

SYN (4100); SYN,ACK (900, 4101); ACK (4101, 901).

(b) As on the slide, segments are numbered one by one. The sender sends 70 to 75; segment 72 is lost. Give the ACK the receiver returns for each segment, when Reno retransmits, and the ACK after 72 arrives. (3 pts)

70: 71; 71: 72; 73: 72; 74: 72; 75: 72. The third duplicate ACK 72 triggers the fast retransmit of 72; when it arrives, ACK **76** (73 to 75 were already received).

Marking. (a) about 0.7 per segment. (b) ACK list 1.5, moment 0.75, ACK 76 0.75.

6. Session 6: e-mail (5 points)

mirrors hta6p p. 4 and 7 to 9 (architecture, SMTP, POP3, IMAP)

(a) A message goes from Alice's mail client to Bob's mailbox and Bob reads it. Name the agents in order and the protocol at each step. How does the sending MTA find Bob's domain's MTA? (2.5 pts)

MUA (submission) → MTA, SMTP → MTA of Bob's domain, SMTP → MDA (mailbox) → Bob's MUA, POP3 or IMAP. MX record in DNS.

(b) Give the ports of SMTP (server to server and submission), POP3 and IMAP. (1 pt)

SMTP 25, submission 465 (formerly 587); POP3 110, 995 with TLS; IMAP 143, 993 with TLS.

(c) Why is IMAP better than POP3 for reading mail on two devices? (1.5 pts)

IMAP queries the message database on the server (folders, browse without retrieving); POP3 only retrieves the inbox, optionally deleting it from the server.